SUMMER CAMP — ASSIGNMENT 5

C Programming — Arrays & String Error Detection

Submitted By

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Total Problems	5 Programs
Assignment	Assignment 5 — Arrays & String Error Detection

ASSIGNMENT OVERVIEW

Problem	Focus Area	Error Types	Count
P1	Array out-of-bounds, invalid index types, integer division, format specifier, array assignment	Syntax, Semantic, Runtime, Logical	**16**
P2	Off-by-one, logical errors, buffer overflow, invalid index types, missing semicolon, array assignment	Syntax, Semantic, Runtime, Logical	**20**
P3	Logical errors in finding lowest, off-by-one, buffer overflow, scanf with array, array assignment	Syntax, Semantic, Runtime, Logical	**19**
P4	Logical errors in max/average, buffer overflow, invalid indices, missing semicolons, null terminator	Syntax, Semantic, Runtime, Logical	**21**
P5	Comprehensive array errors — logical, runtime, semantic, and syntax across all common patterns	All types	**22**

Problem 1 — Array Basics — Marks & Student Data

Description:

Covers fundamental array errors using a student marks system: off-by-one in loops, writing beyond array bounds, float/string as array index, integer division for float result, wrong format specifier, missing & in scanf, direct array assignment, and char array index used with string literal.

Code (with intentional errors):

```
int marks[5]={80,90,70,60,50}  // missing semicolon

for(int i=0;i<=5;i++)         // i<=5 reads marks[5] — out-of-bounds

average = total / 5;           // integer division loses decimal

printf("Average = %d", average); // float printed with %d

scores[5] = 60;                // out-of-bounds on scores[5]

strcpy(subject, "Programming"); // 11 chars in subject[5]

attendance["one"] = 1;         // string index

attendance[2.5] = 1;           // float index

for(int i=0;i<=5;i++) attendance // off-by-one

scanf("%d", arr[i]);           // missing & — should be &arr[i]

grade[0] = "A";                // string literal to char element

if(prices[2] = 300)            // assignment instead of ==

inventory = prices;            // array to array direct assignment

printf("%d", salary[0]);       // float printed with %d

city = "Jaipur";               // string assigned to char array

for(int i=0;i<=5;i++) result   // off-by-one
```

Error Analysis — 16 Errors Found:

Line No.	Error Type	Error Description	Correction
6	**Syntax**	Missing semicolon after array initializer 'int marks[5] = {...}'	*int marks[5] = {80,90,70,60,50};*
12	**Runtime**	Loop i<=5 reads marks[5] on a size-5 array — out-of-bounds access	*for(int i=0; i<5; i++)*
20	**Logical**	average = total/5 performs integer division; decimal part lost for float variable	*average = (float)total / 5;*
22	**Semantic**	printf("%d", average) — float variable printed with integer format %d	*printf("Average = %.2f\n", average);*
31	**Runtime**	scores[5] = 60 writes to index 5 on scores[5] — out-of-bounds (valid: 0-4)	*Declare scores[6] or don't assign beyond index 4*
41	**Runtime**	strcpy of 11 chars into subject[5] — buffer overflow	*char subject[15]; strcpy(subject, "Programming");*
45	**Semantic**	attendance["one"] — string used as array index; must be integer	*attendance[0] = 1;*
47	**Semantic**	attendance[2.5] — float used as array index; must be integer	*attendance[2] = 1;*
54	**Runtime**	for(i<=5) loop on attendance[5] accesses attendance[5] — out-of-bounds	*for(int i=0; i<5; i++)*
		scanf("%d", arr[i]) — missing &	

60	**Semantic**	operator; should pass address of arr[i]	*scanf("%d", &arr[i]);*
72	**Semantic**	grade[0] = "A" — assigns string literal (char*) to char element; needs single quotes	*grade[0] = 'A';*
77	**Logical**	if(prices[2] = 300) — assignment instead of comparison; always true	*if(prices[2] == 300)*
82	**Semantic**	inventory = prices — cannot assign array to array directly in C	*Use memcpy(inventory, prices, sizeof(prices)); or loop copy*
88	**Semantic**	printf("%d", salary[0]) — float element printed with %d format	*printf("%.2f\n", salary[0]);*
93	**Semantic**	city = "Jaipur" — cannot assign string to char array with =; use strcpy()	*strcpy(city, "Jaipur");*
98	**Runtime**	for(i<=5) on result[5] writes result[5] — out-of-bounds	*for(int i=0; i<5; i++)*

Problem 2 — Array Operations — Numbers & Fees

Description:

Array operations program with errors in loop bounds, sum accumulation, finding highest mark (wrong condition), out-of-bounds attendance write, buffer overflow in strings, invalid index types, missing & in scanf, array-to-array assignment, and wrong format specifier for float.

Code (with intentional errors):

```
for(int i=0;i<=10;i++) numbers // off-by-one

sum = numbers[i];              // replaces instead of accumulates

average = sum / 10;            // integer division

printf("Average = %d", average); // float with %d

if(marks[i] < highest)         // logical: finds LOWEST not highest

attendance[5]=1 on attendance[5] // out-of-bounds

for(int i=0;i<=5;i++) present  // off-by-one

strcpy(subject, "ComputerScience"); // 15 chars in subject[10]

branch = "IT";                 // array assignment with =

college not null-terminated    // no '\0' after college[4]

fees["two"] = 50000;           // string index

fees[3.5] = 60000;             // float index

scanf("%d", result[i]);        // missing &

if(stock[2] = 300)             // assignment not ==

backup = stock;                // array-to-array assignment

printf("%d", salary[0]);       // float with %d

for(i<=5) inventory            // off-by-one

strcpy(city,"JaipurCity");     // 10 chars in city[8]

count = inventory;             // array assignment

printf("End...")               // missing semicolon
```

Error Analysis — 20 Errors Found:

Line No.	Error Type	Error Description	Correction
16	**Runtime**	Loop i<=10 reads numbers[10] on size-10 array — off-by-one out-of-bounds	*for(int i=0; i<10; i++)*
22	**Logical**	sum = numbers[i] replaces sum each iteration instead of accumulating	*sum += numbers[i];*
27	**Logical**	average = sum/10 performs integer division; decimal result lost	*average = (float)sum / 10;*
29	**Semantic**	printf("%d", average) — float printed with integer format %d	*printf("Average = %.2f\n", average);*
39	**Logical**	if(marks[i] < highest) updates highest when smaller found — finds LOWEST not highest	*if(marks[i] > highest) highest = marks[i];*
52	**Runtime**	attendance[5]=1 writes to index 5 on attendance[5] — out-of-bounds	*Declare attendance[6] or don't write to index 5*
57	**Runtime**	Loop i<=5 on attendance[5] reads attendance[5] — out-of-bounds	*for(int i=0; i<5; i++)*
63	**Runtime**	strcpy of 15-char string into subject[10] — buffer overflow	*char subject[20]; strcpy(subject, "ComputerScience");*
69	**Semantic**	branch = "IT" — cannot assign string literal to char array with =	*strcpy(branch, "IT");*
76	**Runtime**	college[0..4] set but no null terminator — printf("%s") prints	*Add: college[5] = '\0'; or declare college[6]*

		garbage beyond	
80	**Semantic**	fees["two"] — string used as array index; must be integer	*fees[1] = 50000;*
82	**Semantic**	fees[3.5] — float used as array index; must be integer	*fees[3] = 60000;*
91	**Semantic**	scanf("%d", result[i]) — missing & operator for scanf address argument	*scanf("%d", &result[i]);*
100	**Logical**	if(stock[2] = 300) — assignment always true, not comparison	*if(stock[2] == 300)*
105	**Semantic**	backup = stock — cannot copy array with = in C	*memcpy(backup, stock, sizeof(stock));*
110	**Semantic**	printf("%d", salary[0]) — float element printed with %d format	*printf("%.2f\n", salary[0]);*
115	**Runtime**	Loop i<=5 on inventory[5] writes inventory[5] — out-of-bounds	*for(int i=0; i<5; i++)*
121	**Runtime**	strcpy(city, "JaipurCity") — 10 chars + null into city[8] — buffer overflow	*char city[15]; strcpy(city, "JaipurCity");*
126	**Semantic**	count = inventory — cannot assign array to array with =	*memcpy(count, inventory, sizeof(inventory));*
131	**Syntax**	Missing semicolon after final printf statement	*printf("Program Ended\n");*

Problem 3 — Student Result Analysis

Description:

Student result analysis program. Key errors: integer division for average, wrong condition for finding lowest value, off-by-one attendance loop, buffer overflows, scanf with wrong format for arrays, direct array assignment, invalid index types, and missing return semicolon.

Code (with intentional errors):

```
average = total / 5;            // integer division

if(marks[i] > lowest)           // should be < to find lowest

for(int i=0;i<=5;i++) present  // off-by-one on attendance[5]

strcpy(subject[8],"Programming") // 11 chars in subject[8]

scanf("%s", &branch)             // & not needed for char array

fees[5] = 100000;                // out-of-bounds on fees[5]

if(stock[2] = 30)                // assignment not ==

backup = stock;                  // array assignment

printf("%d", salary[0]);         // float with %d

scanf("%d", result[i]);          // missing &

city = "Jaipur";                 // array assignment with =

inventory["one"]=100;            // string index

inventory[2.5]=200;              // float index

for(i<=5) inventory              // off-by-one

college[0..4] no null term       // missing '\0'

count = inventory;               // array assignment

strcpy(course[6],"Computer");    // 8 chars in course[6]

for(i<=5) numbers                // off-by-one

return 0                         // missing semicolon
```

Error Analysis — 19 Errors Found:

Line No.	Error Type	Error Description	Correction
21	**Logical**	average = total/5 — integer division truncates decimal for float result	*average = (float)total / 5;*
35	**Logical**	if(marks[i] > lowest) updates lowest when LARGER found — finds highest, not lowest	*if(marks[i] < lowest) lowest = marks[i];*
51	**Runtime**	Loop i<=5 on attendance[5] reads attendance[5] — out-of-bounds	*for(int i=0; i<5; i++)*
58	**Runtime**	strcpy of 11 chars into subject[8] — buffer overflow	*char subject[15]; strcpy(subject, "Programming");*
63	**Semantic**	scanf("%s", &branch) — char array already decays to pointer; & is incorrect here	*scanf("%s", branch);*
72	**Runtime**	fees[5] = 100000 writes to index 5 on fees[5] — out-of-bounds (valid: 0-4)	*Declare fees[6] or don't assign index 5*
77	**Logical**	if(stock[2] = 30) — assignment instead of comparison; always true	*if(stock[2] == 30)*
82	**Semantic**	backup = stock — cannot assign arrays with = in C	*memcpy(backup, stock, sizeof(stock));*
87	**Semantic**	printf("%d", salary[0]) — float printed with integer format	*printf("%.2f\n", salary[0]);*
92	**Semantic**	scanf("%d", result[i]) — missing & for scanf address	*scanf("%d", &result[i]);*

99	**Semantic**	city = "Jaipur" — cannot assign string to char array with =	*strcpy(city, "Jaipur");*
103	**Semantic**	inventory["one"] — string used as array index	*inventory[0] = 100;*
105	**Semantic**	inventory[2.5] — float used as array index	*inventory[2] = 200;*
112	**Runtime**	Loop i<=5 on inventory[5] accesses inventory[5] — out-of-bounds	*for(int i=0; i<5; i++)*
115	**Runtime**	college[0..4] written but no null terminator added — printf prints garbage	*college[5] = '\0'; or declare char college[6]*
120	**Semantic**	count = inventory — cannot copy array with =	*memcpy(count, inventory, sizeof(inventory));*
124	**Runtime**	strcpy(course, "Computer") — 8 chars + null into course[6] — buffer overflow	*char course[10]; strcpy(course, "Computer");*
130	**Runtime**	Loop i<=5 on numbers[5] writes numbers[5] — out-of-bounds	*for(int i=0; i<5; i++)*
~last	**Syntax**	Missing semicolon after 'return 0'	*return 0;*

Problem 4 — Sales Report System

Description:

Sales report system with array-based data. Errors: wrong comparison for highest sale (finds lowest), integer division for average, off-by-one loops, buffer overflows, string/float array indices, missing & in scanf, array assignment with =, float printed with %d, null terminator missing, and wrong format specifier.

Code (with intentional errors):

```
averageSales = totalSales/5;   // integer division

printf("Average = %d", avg);    // float with %d

if(sales[i] < highestSale)      // finds LOWEST not highest

monthlySales[5] = 250;          // out-of-bounds on [5]

for(i<=5) monthlySales          // off-by-one

strcpy(product[8],"LaptopComputer") // 14 chars in [8]

category = "Electronics";       // array assignment

stock["first"] = 100;           // string index

stock[2.7] = 200;               // float index

scanf("%d", orders[i]);         // missing &

if(revenue[2] = 3000)           // assignment not ==

backupRevenue = revenue;        // array assignment

printf("%d", commission[0]);    // float with %d

city = "Jaipur";                // array assignment

branch[0..3] no null term       // missing '\0'

for(i<=5) employees             // off-by-one

strcpy(department[6],"Marketing") // 9 chars in [6]

expenses = sales;               // array assignment

ratings[-1] = 5;                // negative index

printf("%s", feedback[0]);      // int with %s
```

```
printf("Report...")             // missing semicolon
```

Error Analysis — 21 Errors Found:

Line No.	Error Type	Error Description	Correction
20	**Logical**	averageSales = totalSales/5 — integer division; decimal lost for float result	*averageSales = (float)totalSales / 5;*
22	**Semantic**	printf("%d", averageSales) — float printed with %d format	*printf("Average Sales = %.2f\n", averageSales);*
29	**Logical**	if(sales[i] < highestSale) updates when smaller — this finds LOWEST not highest	*if(sales[i] > highestSale) highestSale = sales[i];*
41	**Runtime**	monthlySales[5] = 250 writes to index 5 on monthlySales[5] — out-of-bounds	*Declare monthlySales[6] to hold 6 elements*
45	**Runtime**	Loop i<=5 on monthlySales[5] accesses index 5 — out-of-bounds	*for(int i=0; i<5; i++)*
50	**Runtime**	strcpy of 14-char string into product[8] — buffer overflow	*char product[20]; strcpy(product, "LaptopComputer");*
55	**Semantic**	category = "Electronics" — cannot assign string to array with =	*strcpy(category, "Electronics");*
60	**Semantic**	stock["first"] — string literal used as array index; must be integer	*stock[0] = 100;*
62	**Semantic**	stock[2.7] — float used as array index; must be integer	*stock[2] = 200;*
71	**Semantic**	scanf("%d", orders[i]) — missing & address operator	*scanf("%d", &orders[i]);*

80	**Logical**	if(revenue[2] = 3000) — assignment always evaluates to 3000 (true), not comparison	*if(revenue[2] == 3000)*
85	**Semantic**	backupRevenue = revenue — cannot copy array with = assignment in C	*memcpy(backupRevenue, revenue, sizeof(revenue));*
90	**Semantic**	printf("%d", commission[0]) — float element printed with integer format %d	*printf("%.2f\n", commission[0]);*
95	**Semantic**	city = "Jaipur" — cannot assign string to char array with =	*strcpy(city, "Jaipur");*
100	**Runtime**	branch[0..3] written but no null terminator — printf("%s") prints garbage after	*branch[4] = '\0'; or declare branch[5]*
106	**Runtime**	Loop i<=5 on employees[5] accesses employees[5] — out-of-bounds	*for(int i=0; i<5; i++)*
112	**Runtime**	strcpy of 9-char string into department[6] — buffer overflow	*char department[15]; strcpy(department, "Marketing");*
117	**Semantic**	expenses = sales — cannot assign one array to another with =	*memcpy(expenses, sales, sizeof(sales));*
122	**Runtime**	ratings[-1] = 5 — negative index accesses memory before the array	*Use valid indices ratings[0]..ratings[4]*
127	**Semantic**	printf("%s", feedback[0]) — integer element printed with string format %s	*printf("%d\n", feedback[0]);*

~last	**Syntax**	Missing semicolon after final printf("Report Generated Successfully")	*printf("Report Generated Successfully\n");*

Problem 5 — Student Performance Analysis

Description:

A comprehensive student performance analysis program combining all common array error patterns: wrong comparison for highest/lowest, off-by-one bounds, out-of-bounds writes, buffer overflow, invalid indices, array assignment, missing & in scanf, wrong format specifiers, null terminator missing, and negative index.

Code (with intentional errors):

```
average = total / 10;          // integer division

printf("Average = %d", average); // float with %d

if(marks[i] < highest)           // finds LOWEST not highest

if(marks[i] > lowest)            // finds HIGHEST not lowest

attendance[10]=0 on [10]         // out-of-bounds

for(i<=10) present               // off-by-one

strcpy(subject[10],"DataStructures") // 14 chars in [10]

department = "InformationTechnology" // array assignment

fees["two"]=50000;               // string index

fees[4.5]=70000;                 // float index

scanf("%d", result[i]);          // missing &

if(inventory[2] = 30)            // assignment not ==

backup = inventory;              // array assignment

printf("%d", salary[0]);         // float with %d

city = "Jaipur";                 // array assignment

branch[0..3] no null term        // missing '\0'

for(i<=5) stock                  // off-by-one

strcpy(course[8],"Programming")  // 11 chars in [8]

ratings[-1]=10;                  // negative index

printf("%s", feedback[0]);       // int with %s
```

```
scores = marks;                  // array assignment (marks[10] -> scores[5])

printf("%d", temp[5]);           // out-of-bounds on temp[5]
```

Error Analysis — 22 Errors Found:

Line No.	Error Type	Error Description	Correction
20	**Logical**	average = total/10 — integer division truncates decimal part for float	*average = (float)total / 10;*
22	**Semantic**	printf("%d", average) — float printed with integer format %d	*printf("Average = %.2f\n", average);*
29	**Logical**	if(marks[i] < highest) updates when smaller — finds LOWEST instead of highest	*if(marks[i] > highest) highest = marks[i];*
39	**Logical**	if(marks[i] > lowest) updates when larger — finds HIGHEST instead of lowest	*if(marks[i] < lowest) lowest = marks[i];*
51	**Runtime**	attendance[10]=0 writes to index 10 on attendance[10] — out-of-bounds (valid 0-9)	*attendance[9] = 0; or declare attendance[11]*
55	**Runtime**	Loop i<=10 on attendance[10] reads index 10 — off-by-one out-of-bounds	*for(int i=0; i<10; i++)*
62	**Runtime**	strcpy of 14-char string into subject[10] — buffer overflow	*char subject[20]; strcpy(subject, "DataStructures");*
68	**Semantic**	department = "InformationTechnology" — cannot assign string to char array with =	*strcpy(department, "InformationTechnology"); // also increase array size to [25]*
73	**Semantic**	fees["two"] — string used as array index; must be integer	*fees[1] = 50000;*
75	**Semantic**	fees[4.5] — float used as array	*fees[4] = 70000;*

		index; must be integer	
84	**Semantic**	scanf("%d", result[i]) — missing & address operator for scanf	*scanf("%d", &result[i]);*
96	**Logical**	if(inventory[2] = 30) — assignment (always true) instead of comparison	*if(inventory[2] == 30)*
101	**Semantic**	backup = inventory — cannot copy array to array with = in C	*memcpy(backup, inventory, sizeof(inventory));*
106	**Semantic**	printf("%d", salary[0]) — float printed with integer format %d	*printf("%.2f\n", salary[0]);*
111	**Semantic**	city = "Jaipur" — cannot assign string to char array with = operator	*strcpy(city, "Jaipur");*
116	**Runtime**	branch[0..3] assigned but no null terminator — printf("%s") prints garbage	*branch[4] = '\0'; or declare branch[5]*
120	**Runtime**	Loop i<=5 on stock[5] writes stock[5] — off-by-one out-of-bounds	*for(int i=0; i<5; i++)*
126	**Runtime**	strcpy of 11 chars into course[8] — buffer overflow	*char course[15]; strcpy(course, "Programming");*
132	**Runtime**	ratings[-1] = 10 — negative index accesses memory before array start	*Use ratings[0]..ratings[4] only*
136	**Semantic**	printf("%s", feedback[0]) — integer printed with string format %s	*printf("%d\n", feedback[0]);*
		scores = marks — cannot assign	

140	**Semantic**	arrays with = (also sizes differ: marks[10] vs scores[5])	*memcpy(scores, marks, sizeof(scores)); // copies only first 5*
146	**Runtime**	printf("%d", temp[5]) — accesses index 5 on temp[5] — out-of-bounds	*temp[4] is the last valid index*

— End of Assignment 5 —